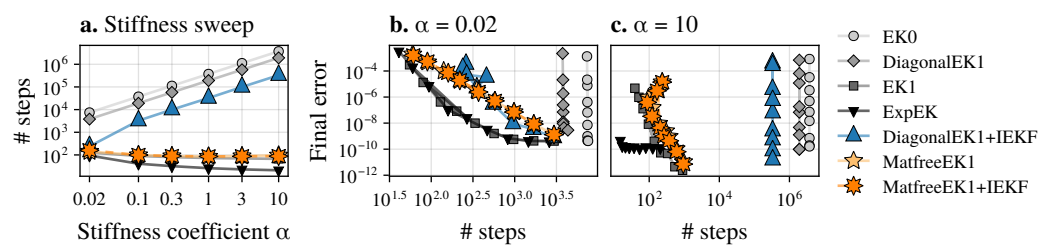
a. Stiffness sweep
steps
10^6
10^5
10^4
10^3
10^2
0.02
0.1
0.3
1
3
10
Stiffness coefficient α
b. α = 0.02
Final error
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
$10^{1.5}$
$10^{2.0}$
$10^{2.5}$
$10^{3.0}$
$10^{3.5}$
steps
c. α = 10
10^2
10^4
10^6
steps
EK0
DiagonalEK1
EK1
ExpEK
DiagonalEK1+IEKF
MatfreeEK1
MatfreeEK1+IEKF